

COMP3004 Group 11 Project

2023-12-13

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Source Code:

<https://github.com/vidun-jay/3004-team-project>

Use Case 1: Applying AED+ (Base Use Case)

Primary Actor:

- Rescuer

Scope:

- Emergency Aid

Level:

- User Goal

Stakeholders and Interests:

- **Rescuer:** Provide emergency treatment to victims with cardiopulmonary arrest to ensure they are out of danger.
- **Victims:** The rescuer used AED+ to perform first aid and saved his life from danger.
- **AED+:** Ensure that it is in normal working condition, can be used by rescuers, and can provide emergency treatment to the rescued person.

Pre-Conditions:

- Victims in cardiac arrest require emergency assistance.
- The AED+ is in good condition and ready for use.
- The rescuer can operate the AED+ and perform the rescue according to the instructions.

Main Success Scenario:

1. Remove the lid of AED+ and turn on the AED+.
2. Follow the audio and visual prompts through each stage of the live separate process.
3. Try to call the patient to see if they are still conscious.
4. Call for help.
5. Look for the rise and fall of the victim's chest and listen for effective breathing coming from the nose or mouth.
6. Use the scissors from the ready kit to cut the victim's clothing to expose their bare chest.
7. Tear open the electrode pack and unfold the electrode pads.
8. Using the sensors crosshairs to guide to align the CPR landmark between the victim's nipples and the middle of their breastbone.
9. Press the CPR landmark with your right hand and pull the number-2 tab to peel the protective backing from the electrode.
10. Repeat the action on the lower pad marked with the number-3 tab.
11. Once the pads are applied, the AED+ will auto-analyze the victim's heart rhythm.
12. Don't let anyone touch the victim during the analysis step or the shock delivery step.
13. Sometimes the early stage of the life-support process may have been started before you arrive with the AED+.

Extensions:

- 2a. An LED light will indicate which step the rescuer should focus on.
- 4a. While waiting for rescue, continue to provide first aid to the victim.
- 5a. If desired, the lid of AED+ (PASS) can be placed behind the victim's shoulders to help support and maintain an open airway.
- 5b. Should not use the lid as a support if you suspect the victim has a spinal injury.
- 6a. If the ready kit is lost, find an alternative way to remove the victim's clothing safely

- 7a. The CPR D pads should be one-piece pads.
- 8a. Please confirm the direction and front and back of the CPR pad.
- 9a. Press the electrode from the center out to make sure it adheres properly to the victim's skin.
- 10a. If the victim is large, you must tear away the lower at the perforated line and extend it before pulling the number-3 tab.
- 10b. Instructions on the electrode pack remind you how to attach the electrodes to the victim.
- 11a. You should not touch the victim at this point.
- 12a. In this case, you can attach the electrode pads to the chest before switching on the AED.
- 12b. AED+ will jump straight to the analysis stage to save time.
- 12c. This enables you to delete therapy and CPR treatment faster.

Postconditions:

The AED+ was placed correctly, the rescuer successfully operated the AED+ as instructed, and the victims received the correct shock.

Use Case 2: Real-Time CPR

Actors:

- Rescuer
- Victim

Stakeholders and Interests:

- Rescuer: Trying to improve the chances of survival for the victim with the help of the AED+.
- Victim: Wanting to have their lives saved by the rescuer and the AED+.

Preconditions:

- AED+ is in good condition, operational, and turned on
- Real CPR Help is enabled
- The victim is an adult who requires CPR

Main Success Scenario:

1. The rescuer starts giving CPR to the victim with the guidance of the AED+ if no shock is advised.
2. AED+ produces a metronomic beep to indicate to the rescuer the rate of compressions.
3. AED+ will display visual prompts and audio feedback to instruct the rescuer on the necessary depth of compression.
4. Rescuer adjusts the compressions based on the feedback it gets from AED+.
5. Periodically check the condition of the victim.

Extensions:

- 2a. The rescuer isn't timing the compressions with the metronomic beep that the ED+ is providing
 - AED+ either increases or decreases the rate of beep to instruct the rescuer to either do them faster or slower.
- 3a. Compressions aren't deep enough
 - AED+ will display a "PUSH HARDER" message to tell the rescuer to go deeper in the compressions
- 3b. Rescuer is too fatigued to make proper compressions
 - The rescuer passes the job to another person for them to listen to the AED+ and make proper chest compressions.

Postconditions:

- Victim is receiving proper CPR, AED+ continues to be used until professional help arrives

Use Case 3: AED+ Deliver shock to Victim**Primary Actor:**

- AED+

Scope:

- AED+ feature

Stakeholders and Interests:

- AED+ delivers shock to the victim's heart according to the vital signs and gives it heart recovery function

Pre-Conditions:

- AED+ has been correctly applied to the victim.
- The victim is advised to be shocked.

Main Success Scenario:

1. The rescuer follows the instruction of the AED+ Audio system to apply electrodes to the victim's chest.
2. AED+ will auto-analyze the heart rhythm.
3. AED+ will deliver a shock to the victim if he/she is shockable.
4. After each shock is delivered, AED+ will inform the rescuer by the Audio system to do CPR.
5. AED+ will cycle from step 2 to step 4 until the victim's heart rate reaches a stable state.

Extensions:

- 2a. If the victim's heart rhythm is not shockable, the AED+ will use an Audio system to inform the rescuer to do the CPR.

Postconditions:

The victim's heart rate reaches a stable state.

Use Case 4: AED+ Self-Test**Actors:**

- AED+
- Maintenance personnel

Stakeholders and Interests:

- AED+: Must properly perform its self-test to ensure reliability and proper functionality when in use.
- Maintenance personnel: Performs routine checks on the AED+ to ensure it's in working condition and address any issues that come up during the self-test.

Preconditions:

- AED+ is in good condition, operational, and turned on.

Main Success Scenario:

1. Once the AED+ is on and at scheduled intervals, it performs an automatic self-test.
2. AED+ completes its self-test and finds no issues. It then displays a green check mark on the display.

Extensions:

2a. A red X was displayed after the tests

- Maintenance personnel are alerted.
- They troubleshoot the AED+ to fix the issue that caused the self-test to fail.
- They perform checks to ensure AED+ is fully functional after making the necessary maintenance.

Postconditions:

- AED+ has been maintained and is in working order

Use Case 5: Safety: AED+ Battery

Actors:

- AED+
- Maintenance personnel

Scope:

- Safety feature

Stakeholders and Interests:

- The AED+ will consume power during storage and use. The battery should be checked and replaced in time to ensure that it can operate normally.
- AED+ should be checked monthly to ensure the batteries are in good condition.

Preconditions:

- AED+'s batteries are out of power
- AED+'s batteries are expiration.

Main Success Scenario:

1. Ensure the AED+ is turned off.
2. Open the battery compartment by removing the battery cover from the back of the unit.
3. Remove all batteries at once and dispose of batteries properly.
4. Inserting new batteries.
5. After installing new batteries, press the Battery Reset Button inside the battery compartment when prompted.

Extensions:

- 2a. First, Insert a small tool into the two slots on the rear of the unit to depress the latches.
- 2b. Second, insert the tool into the groove on the bottom to lift the cover
- 4a. Use Duracell, Sanyo, or Varta batteries only.
- 4b. Do not use Panasonic or Rayovac batteries.
- 4c. You MUST replace all 10 batteries at once. Do not replace individual batteries.
- 5a. Pressing the button resets the battery usage indicator to full charge

Postconditions:

- AED+ has been maintained and all batteries are in good condition.

Use Case 6: Safety: Electrode has been Removed

Actors:

- AED+
- Maintenance personnel

Scope:

- Safety feature

Stakeholders and Interests:

- Provide warning while electrode does not connect with AED+ correctly

Preconditions:

- AED+'s electrode does not connect well with AED+

Main Success Scenario:

1. AED+ stops working and displays a warning message.
2. Remove the electrode.
3. Check whether the connected interface is blocked by dust or foreign matter.
4. Clean the connected interface.
5. Re-connect the electrode with AED+.

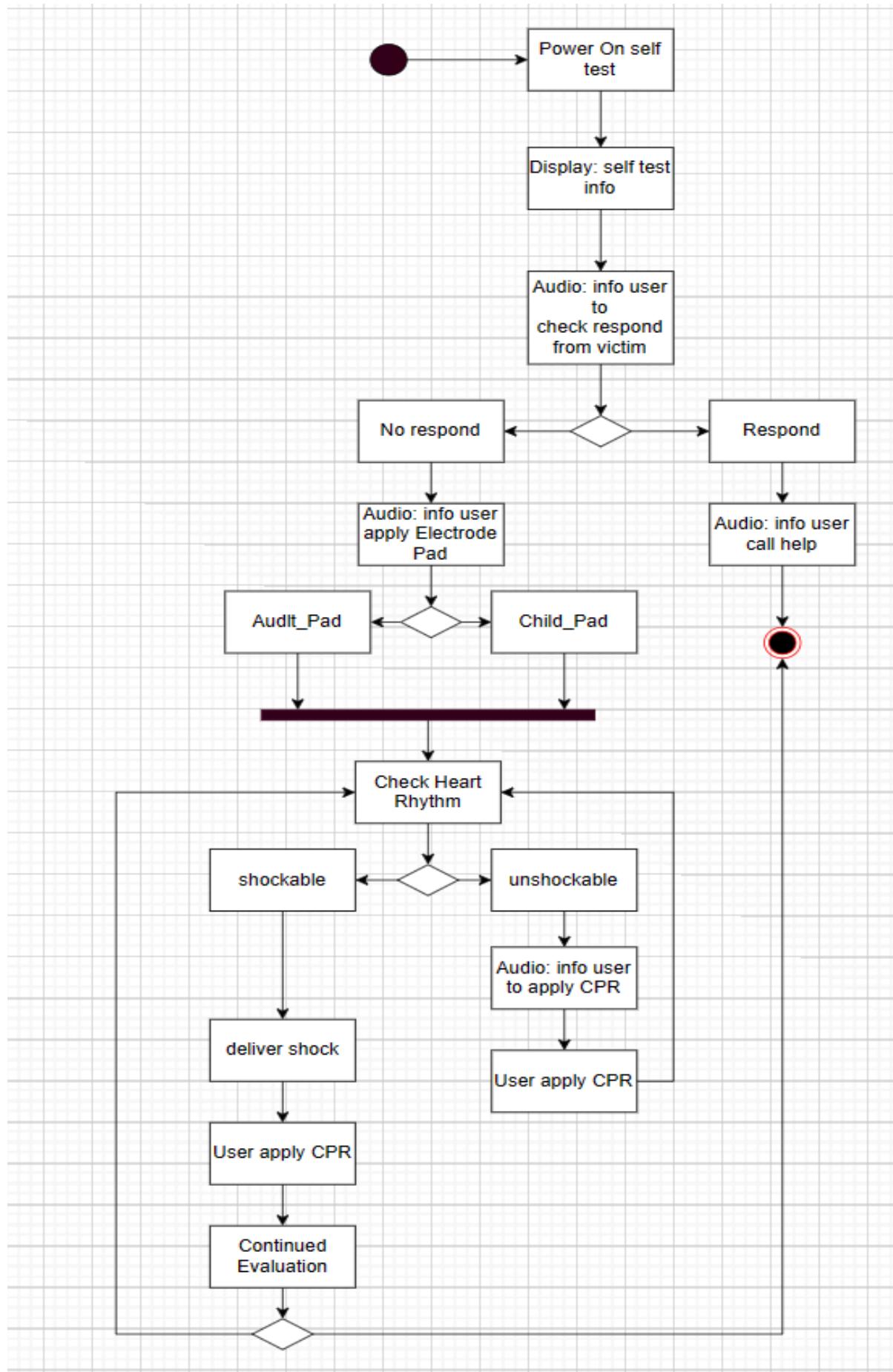
Extensions:

- 1a. This situation happens when the electrode connection either does not connect well or does not connect with the AED+.
- 3a. If the connected interface is damaged, replace it with a new pair.
- 5a. If the warning still appears, use the user's manual to do a manual test.

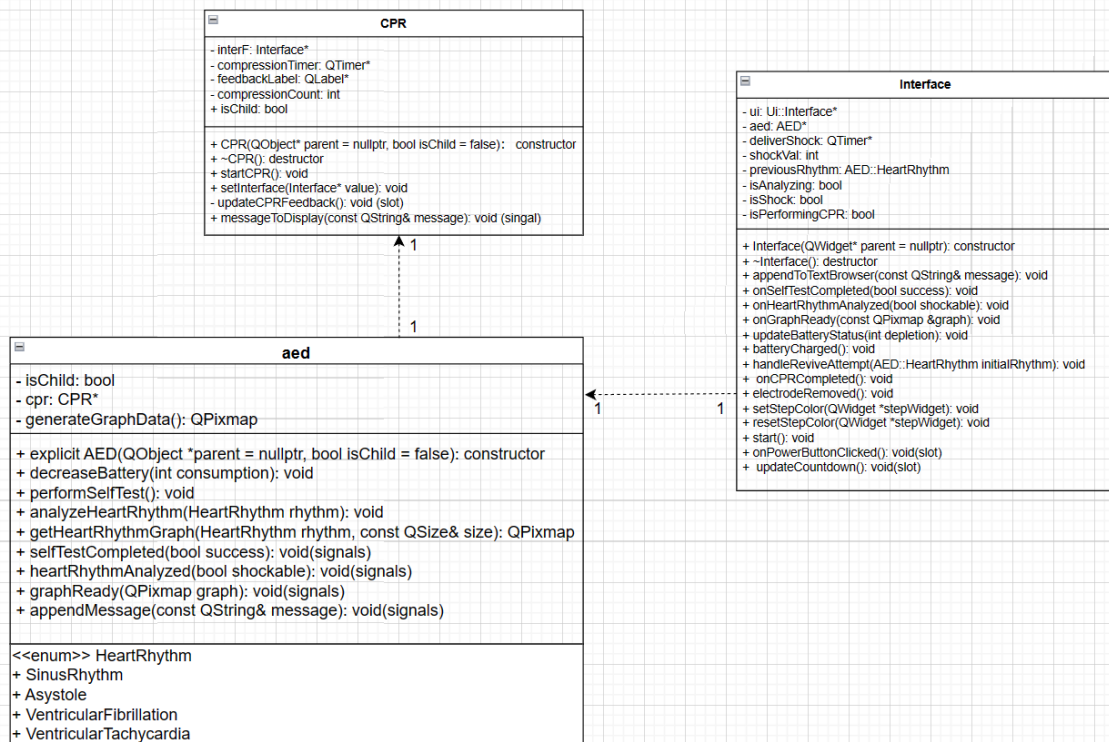
Postconditions:

The rescuer received the warning and checked the connection of the electrode with the AED+.

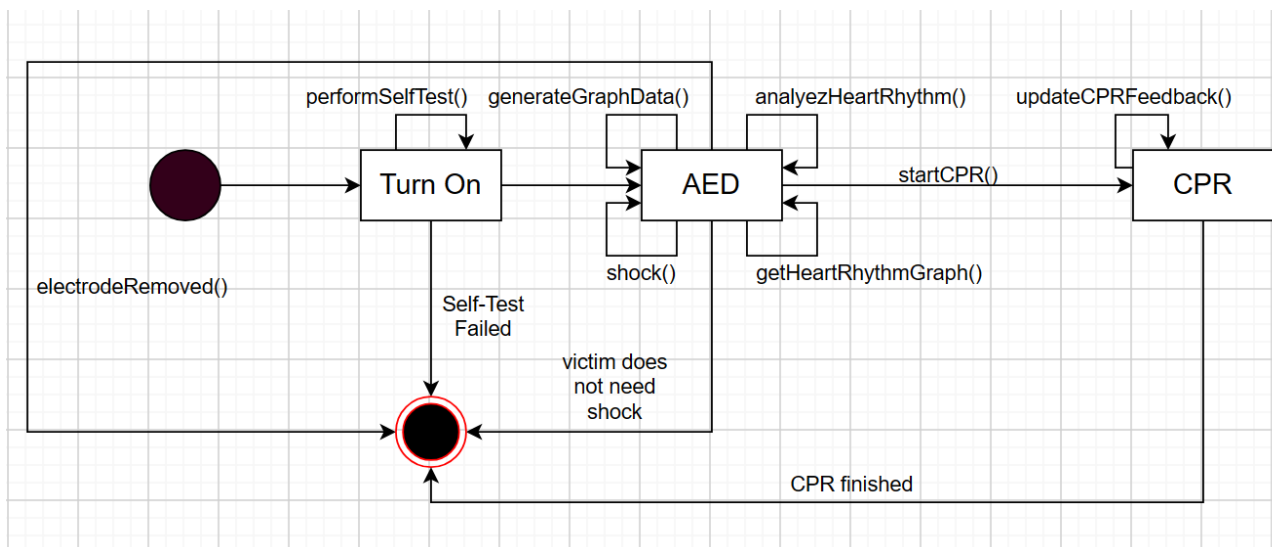
Active Diagram



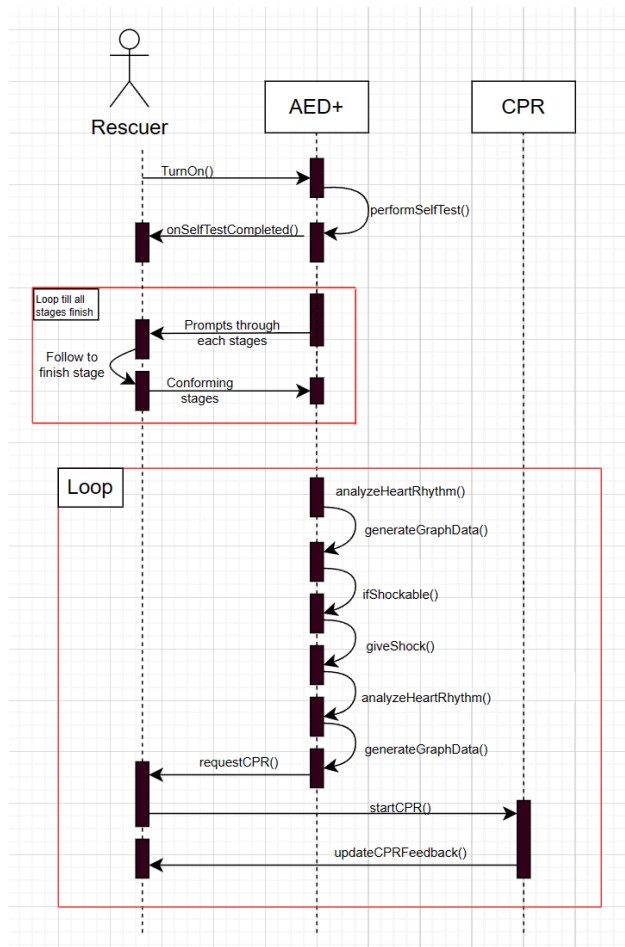
UML Class diagram



State Diagram

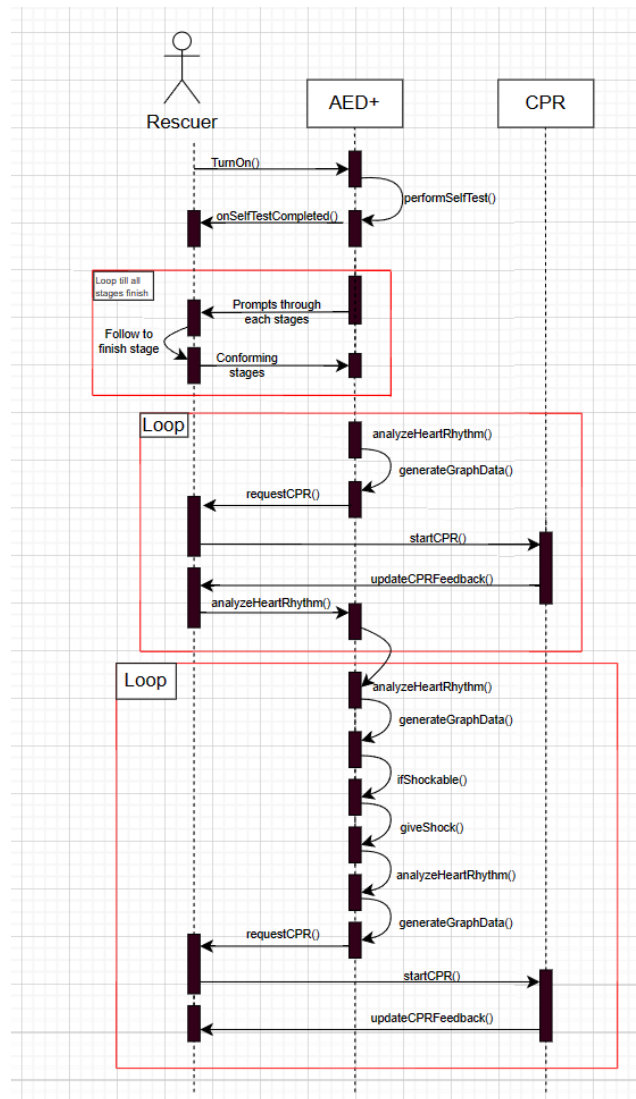


Sequence diagram1: Victim is shockable



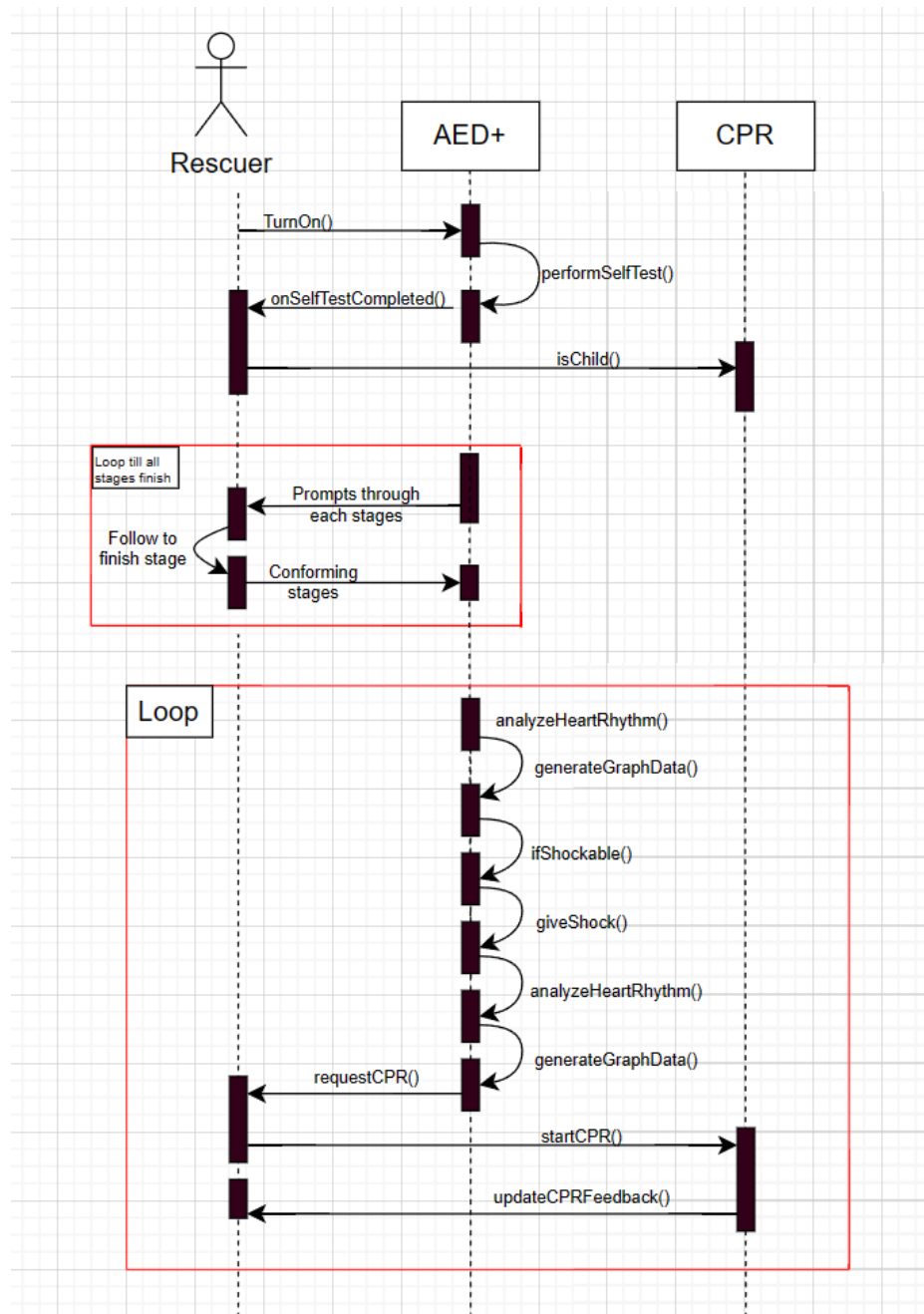
- Responder Turns on AED
- AED performs self-test
- The responder puts electrode pads on the victim automatically
- AED loops through instructions between analyzing the victim's heart rate, shocking(if the heart rhythm is asystole), and requesting CPR
- The heart rhythm can only be Vfib or Vtach if a shock is delivered

Sequence diagram2: Victim is not shockable



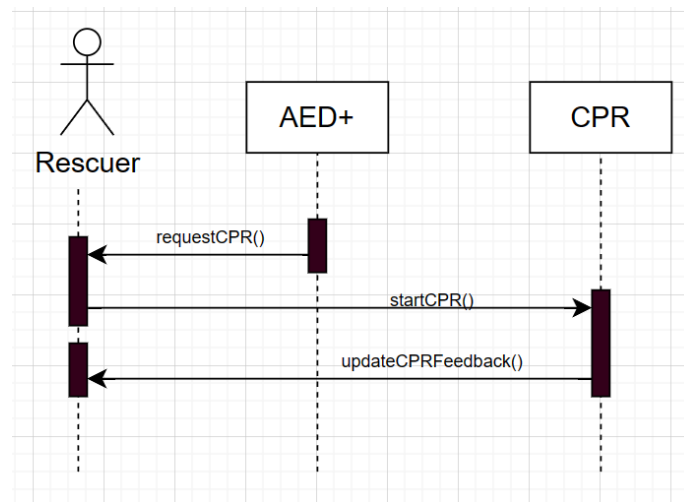
- Responder Turns on AED
 - AED performs self-test
- AED analyzes the heart rhythm of the victim and requests CPR before a shock
 - Responder provides CPR
- AED checks if the victim's heart rhythm is in a shockable state or requests for more CPR
- No shock should be delivered to the victim if their heart rate is either Sinus or Asystole.
Only perform CPR on an Asystole victim

Sequence diagram3: Victim is Child



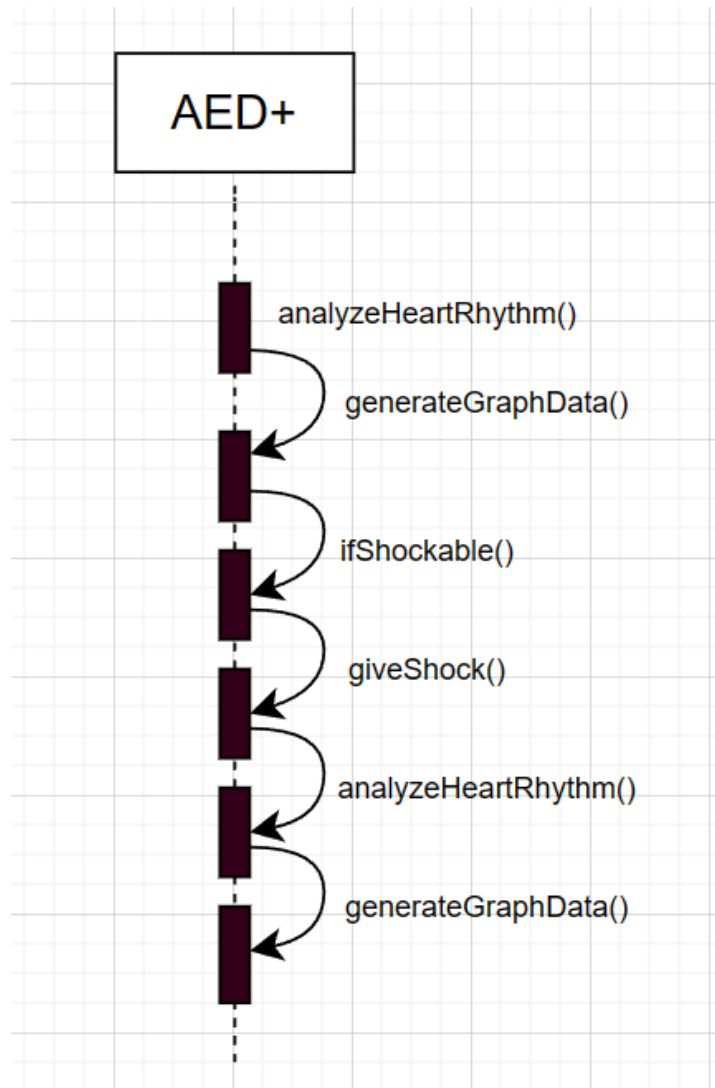
- Same steps as adult victim shockable/non-shockable except ischild() is set to true so the AED knows to use less power

Sequence diagram4: Real-Time CPR



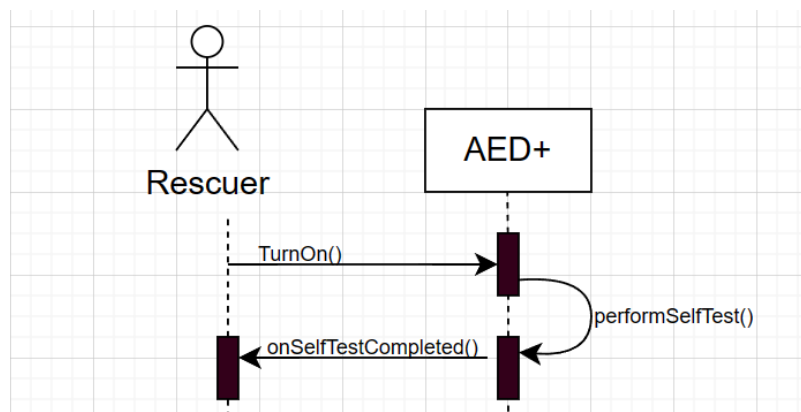
- AED requests the responder to give CPR to the victim
- `updateCPRFeedback` will provide feedback to the responder on their CPR

Sequence diagram5: Deliver shock to victim



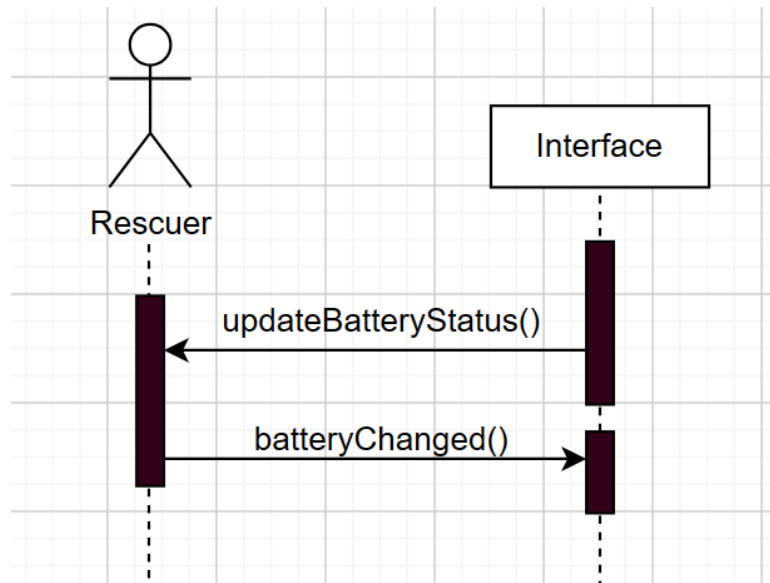
- Analyze the victim's heart rhythm and create a graph
- If heart rhythm is asystole and sinus, skip giveShock() and request the responder to do CPR
- It will skip to the generateGraphData and display the sinus rhythm graph

Sequence diagram6: AED+ Self-test



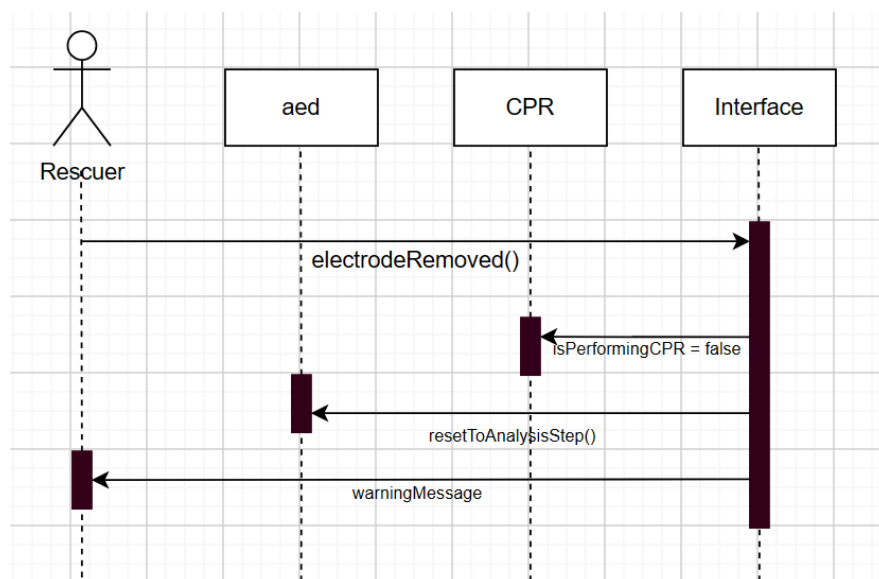
- AED checks to make sure all components are functional
 - Displays test results after completion

Sequence diagram7: Safety: AED+ Battery



- When a power is depleted either through shocks or self tests, the program will update the battery on the display to show the remaining power within the AED
 - Self test: 1% power
 - Adult shock 10% power
 - Child shock 7% power

Sequence diagram8: Safety: Electrode has been Removed



- If an electrode is removed from the victim, sets performingCPR to false
 - Reset victim analysis to step 1 as there is nothing to analyze
- Sends the rescuer a warning message that an electrode was removed

Traceability matrix

ID	Requirement	Related Use Case	Fulfilled By	Test	Description
1	The interface contains all the necessary buttons for operating	N/A	Main, Interface, Interface.ui	In the Qt creator, click run to start.	Interface and Interface.ui is used to provide

	the AED+				users with visual operations on AED+
2	CPR	real-time CPR	CPR.cpp Interface.ui	Start the simulation and the Audio will inform the tester to do the CPR after the analysis and delivery of the shock	The AED+ will determine the heart rhythm if the victim is shockable or not, which would then prompt the rescuer to perform CPR
3	updateCPRFeedback()	real-time CPR	CPR.cpp Interface.ui	Follow the instructions of AED+, it will inform you if you need to push harder.	The program will detect your CPR intensity and remind you when to start, when you need to push harder, or when the pressure is just right.
4	shockable()	Base Use Case, AED+ delivers shock to victim	interface.cpp	Follow the instructions of AED+ systems, the shock will be auto-applied if the victim is shockable	The AED+ will continuously monitor the victim's heart rate and will apply shock when the heart rate is appropriate.
5	performSelfTest()	Base Use Case, AED+ Self-test	aed.cpp	Start the simulation, The self-test information will be performed by AED+	AED+ will automatically detect whether each function can be run when turned on, and will automatically display the test results after completing the test.
6	isChild()	Base Use Case	CPR.cpp Interface.ui	Start simulation and	AED+ has different

				select the 'child' option from the dropdown menu	electrode pads for adults or for children (weight < 40kg)
7	startCPR()	Real-time CPR	CPR.cpp	Start the simulation and click 'Turn on'.	This is to simulate 100 chest compressions/min, if the self-test is passed, it will auto-running.
8	analyzeHeartRhythm()	Base Use Case,	aed.cpp	Click 'Turn on' in the simulation and it will auto-simulate the victim's heart rhythm	This function will simulate analyze and check if the rhythm is shockable or not.
9	getHeartRhythmGraph()	Base Use Case	aed.cpp	Click 'Turn On' and this function will run in the background.	This function generates a QPixmap that represents the heart rhythm graph.
10	generateGraphData()	Base Use Case	aed.cpp, Interface.ui	Click 'Turn On' and you will see the ECC in the Qt interface	After the function getHeartRhythmGraph(), this function will generate the actual graph. And it will randomly generate a victim's ECC.
11	updateBatteryStatus()	Use Case 6: Battery	Interface.cpp	'Turn On' and start the simulation.	When the AED+ is running, all actions will deduct the corresponding power value, which can simulate the consumption of

					the battery during use. AED+ will not be able to turn on or perform functions if the battery is too low.
12	batteryCharged()	Use Case 6: Battery	Interface.cpp	On the UI, click 'recharge battery'	After clicking 'recharge battery', battery replacement will be automatically simulated and the battery indicator will be updated to 100%.
13	electrodeRemoved()	Base Use Case, Real-time CPR, Safety: electrode has been removed	Interface.cpp	On the UI, click 'remove electrode'	When the electrode is removed, the AED+ will issue a prompt and stop running. It will then automatically start back up again at the analysis step (which simulates the user correctly placing the pads back on the victim)

Textual Explanation

Our design choices were driven by specific and practical objectives. We first designed a variety of use cases such as real-time CPR and shock delivery and thought of how many different scenarios the rescuer and victim would be placed in and what sort of help would need to be applied. We then implemented the tools needed for each of our use cases ensuring that the interface was versatile enough to accommodate for all of the different cases.

We considered the different stakeholders involved like rescuers, victims, and maintenance personnel to ensure that our AED+ can meet a broad range of needs in an emergency. For example, regular self-testing features were integrated to give AED+ users and maintenance personnel an easy and reliable way to ensure the device's functionality and ease the troubleshooting process.